

Algebra II with Trigonometry

Chapter 5.3

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Here is a comprehensive breakdown based on the provided document.

1. Short Summary

This document is a guided notes worksheet for an Algebra 2 Trigonometry/Honors class covering Section 5.3: Trigonometric Functions of Angles. It transitions students from right-triangle trigonometry to evaluating trigonometric functions for any angle on the coordinate plane. It introduces how to use x , y , and r coordinates to define the six trigonometric functions, how to determine their signs by quadrant, and how to use reference angles to evaluate them. Additionally, it introduces fundamental trigonometric identities (reciprocal and Pythagorean) and applies trigonometry to find the area of triangles and solve real-world word problems.

2. Core Definitions, Theorems, and Formulas

Core Definitions:

- **Trigonometric Functions on the Coordinate Plane:** Let θ be an angle in standard position and $P(x, y)$ be a point on its terminal side. The distance from the origin to P is $r = \sqrt{x^2 + y^2}$.
- $\sin \theta = \frac{y}{r}$
- $\cos \theta = \frac{x}{r}$
- $\tan \theta = \frac{y}{x} \ (x \neq 0)$
- $\csc \theta = \frac{r}{y} \ (y \neq 0)$
- $\sec \theta = \frac{r}{x} \ (x \neq 0)$

- $\cot \theta = \frac{x}{y}$ ($y \neq 0$)
- **Quadrantal Angles:** Angles whose terminal side lies on the coordinate axes (e.g., 0° , 90° , 180°). Trig functions may be undefined here if x or y is 0.
- **Reference Angle ($\bar{\theta}$):** The acute angle formed by the terminal side of θ and the x -axis.

Theorems & Rules:

- **Signs of Trigonometric Functions by Quadrant:**
- **Quadrant I:** All functions are Positive.
- **Quadrant II:** sin and csc are Positive (others negative).
- **Quadrant III:** tan and cot are Positive (others negative).
- **Quadrant IV:** cos and sec are Positive (others negative).

Fundamental Identities:

- **Reciprocal Identities:**
- $\csc \theta = \frac{1}{\sin \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\cot \theta = \frac{1}{\tan \theta}$
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\cot \theta = \frac{\cos \theta}{\sin \theta}$
- **Pythagorean Identities:**
- $\sin^2 \theta + \cos^2 \theta = 1$
- $\tan^2 \theta + 1 = \sec^2 \theta$
- $1 + \cot^2 \theta = \csc^2 \theta$

Formulas:

- **Area of a Triangle:** $A = \frac{1}{2}ab \sin \theta$ (where a and b are side lengths and θ is the included angle).

3. Worked Study Guide: Step-by-Step Concepts

Concept A: Evaluating a Trig Function for Any Angle To evaluate something like $\tan\left(-\frac{4\pi}{3}\right)$ without a calculator: 1. *Find the reference angle:*

Sketch $-\frac{4\pi}{3}$ (which is -240°). This terminates in Quadrant II. The distance from the terminal side to the x -axis (which is $-\frac{3\pi}{3}$ or -180°) is exactly $\frac{\pi}{3}$ (60°). This is your reference angle. 2. *Determine the sign:* In Quadrant II, only Sine and Cosecant are positive. Therefore, Tangent will be **negative**. 3. *Evaluate:* Calculate the tangent of the reference angle: $\tan(\frac{\pi}{3}) = \sqrt{3}$. Apply the negative sign from step 2. The final answer is $-\sqrt{3}$.

Concept B: Finding Missing Trig Functions Given One Function and a Quadrant

Example text: $\cos \theta = -\frac{3}{5}$, θ in Quadrant II. 1. *Set up x, y, r :* We know $\cos \theta = \frac{x}{r}$. Therefore, $x = -3$ and $r = 5$. (Note: r is a distance and is always positive; the negative sign belongs to the coordinate axis). 2. *Solve for the missing variable:* Use the Pythagorean theorem for the coordinate plane: $x^2 + y^2 = r^2$.

$(-3)^2 + y^2 = (5)^2 \rightarrow 9 + y^2 = 25 \rightarrow y^2 = 16 \rightarrow y = \pm 4$. 3. *Determine the sign of the missing variable:* Because θ is in Quadrant II, the y -coordinates are positive. Therefore, $y = 4$. 4. *Build the remaining functions:* Now that you have $x = -3, y = 4, r = 5$, you can find any function. For example, $\sin \theta = \frac{y}{r} = \frac{4}{5}$.

Concept C: Using the Area Formula To find the area of a non-right triangle, you need "Side-Angle-Side". 1. Identify the two side lengths (a and b). 2. Identify the angle *between* them (θ). 3. Plug into $A = \frac{1}{2}ab \sin \theta$. For example, if sides are 10 and 22, and the included angle is 10° , $A = \frac{1}{2}(10)(22) \sin(10^\circ) = 110 \sin(10^\circ)$.

4. Practice Problems

(These problems are drawn directly from the unworked examples provided in the document).

Problem 1: Find the reference angle for -199° . **Answer:** 19° . *Explanation:* A -199° angle rotates clockwise past the -180° mark (the negative x -axis) by 19° , landing in Quadrant II. The reference angle is the acute angle to the x -axis, which is 19° .

Problem 2: From the information given, find the quadrant that θ lies in: $\tan \theta < 0$ and $\sin \theta < 0$. **Answer:** Quadrant IV. *Explanation:* Tangent is negative in Quadrants II and IV. Sine is negative in Quadrants III and IV. The only quadrant where both are true is Quadrant IV.

Problem 3: Find the values of $\sin \theta$ and $\csc \theta$ if $\cot \theta = -2$ and θ is in Quadrant IV. **Answer:** $\sin \theta = -\frac{\sqrt{5}}{5}$, $\csc \theta = -\sqrt{5}$. *Explanation:* $\cot \theta = \frac{x}{y} = -2$. Since θ is in QIV, x is positive and y is negative, so $x = 2$ and $y = -1$. Find r : $r = \sqrt{2^2 + (-1)^2} = \sqrt{5}$. Therefore, $\sin \theta = \frac{y}{r} = \frac{-1}{\sqrt{5}} = -\frac{\sqrt{5}}{5}$. $\csc \theta$ is the reciprocal of sine, which is $-\sqrt{5}$.

Problem 4: A triangle has an area of 16 in^2 . The angle between two sides of the triangle is 36° , where one of these two sides has a length of 5 in. Find the length of the other side. **Answer:** $b = \frac{32}{5 \sin(36^\circ)}$ inches (approx. 10.89 in). *Explanation:* Use $A = \frac{1}{2}ab \sin \theta$. Substitute knowns: $16 = \frac{1}{2}(5)(b) \sin(36^\circ)$. Multiply by 2: $32 = 5b \sin(36^\circ)$. Divide to isolate b : $b = \frac{32}{5 \sin(36^\circ)}$.

Problem 5: A rocket fired straight upward is tracked by an observer on the ground, one mile away. Show that when the angle of elevation is θ , the height of the rocket (in feet) is $h = 5280 \tan \theta$. **Answer:** *Explanation:* The observer, the launchpad, and the rocket form a right triangle. The distance along the ground (adjacent to the angle of elevation θ) is 1 mile. The height of the rocket is the side opposite θ . Because 1 mile = 5280 feet, the adjacent side is 5280 ft. Using the definition of tangent ($\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$), we get $\tan \theta = \frac{h}{5280}$. Multiplying both sides by 5280 yields $h = 5280 \tan \theta$.

5. Assumptions and Ambiguities Resolved

- **Reference Angle Notation:** The OCR text had missing symbols for the "reference angle," showing up as blank spaces with a line over them. Based on standard mathematical notation and faint visual cues in the PDF, I assumed the document intended to use $\bar{\theta}$ or simply the phrase "reference angle."
- **Coordinate Plane Grids (Example 4):** Example 4 asks to evaluate exact values based on visual coordinate graphs, but the exact points are not

explicitly labeled with numbers in the text. I omitted this specific graphical problem from the practice section to avoid guessing coordinates based on blurry grid lines.

- **Unit Conversion in Word Problem (Example 9):** The text asks to show that the height of a rocket tracked from "one mile away" is $h = 5280 \tan \theta$. The document does not explicitly provide the conversion factor "1 mile = 5280 feet." I assumed this is required prior knowledge and explicitly stated it in the answer key for Practice Problem 5.
- **Shaded Region of a Circle (Example 8c):** The document asks for the area of a shaded region inside a circle. However, the provided formulas only cover the area of a triangle ($A = \frac{1}{2}ab \sin \theta$). Solving this completely would require knowing the formula for the area of a circular sector ($A = \frac{1}{2}r^2\theta$), which is omitted from the provided text. I excluded this specific problem from the study guide and practice section as it cannot be solved *only* with the formulas provided in this PDF.